

FINAL RISK AND CONTINGENCY REGISTER

(RED-TEAM AUDIT)

Project ID: SIH26034
Product: NyayaDrishti-LM
Audience: Entire Engineering Team
Status: FROZEN

Red-Team Threat Matrix & Mitigations

Threat ID	Threat Description	Threat Domain	Impact (1-5)	Probability (1-5)	Risk Score	Real-Time Detection Method	Engineering & Operational Mitigation
TR-01	Monocular Scale Ambiguity Failure: Uncalibrated photo used to allege font height violation in court.	Optical / Legal	5	4	20	ArUco / marker detection returns null or reprojection error > 1.2 px.	Disallow mm calculation; assign REQUIRES_REVIEW (NO_CALIBRATION_TARGET) ; fall back to relative ratio; mandate manual caliper check.
TR-02	Specular Glare on Metallic Packaging: Foil blooming obliterates MRP or consumer care email, causing false missing-declaration flag.	Optical / Vision	4	4	16	Pre-inference HSV saturation filter ($V > 245, S < 15$) intersecting text polygon.	Immediately trigger UNABLE_TO_VERIFY (SPECULAR_GLARE) with interactive viewfinder HUD guiding inspector to tilt camera 15°.
TR-03	Retroactive Law Enforcement: Product packed in 2022 flagged for lacking Unit Sale Price (USP) or 2026 Country of Origin filter.	Statutory / Legal	5	3	15	Manufacturing date parser outputting date earlier than GSR enactment date.	Temporal Statutory Epoch Dispatcher dynamically matches product Mfg Date to historical immutable rule snapshot.
TR-04	Dot-Matrix / Thermal Ink-Jet Coder OCR Error: Fragmented dot characters misread (e.g., ₹48 read as ₹40, causing	AI / OCR	4	3	12	OCR token confidence < 0.85 or discrepancy between price and category baseline.	Morphological closure filter; dual-engine consensus (PaddleOCR + Tesseract v5); route mathematical mismatch to REQUIRES_HUMAN_REVIEW .

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	false USP arithmetic mismatch).						
TR-05	Viral AGPL License Contamination: Ultralytics YOLO code included in build, forcing open-sourcing of government proprietary logic.	IP / Legal	5	2	10	Pre-commit license scanner and dependency audit.	Strict architectural prohibition of Ultralytics YOLO; use Apache-2.0 DBNet++ and RT-DETR.
TR-06	Network Blackout in Wholesale Mandi: Inspection fails because application attempts to call a cloud API.	Deployment / Field	4	4	16	System ping check to external endpoints.	100% offline edge execution via INT8 CPU-quantized ONNX Runtime and local SQLite database; zero network calls.
TR-07	Section 63 BSA Evidence Inadmissibility: Court or magistrate rejects digital report as unauthenticated hearsay.	Quasi-Judicial	5	2	10	Merkle DAG integrity verification check.	Complete SHA-256 Merkle provenance chaining linking raw capture, rectified crop, OCR tokens, and officer digital signature into a tamper-evident PDF dossier.
TR-08	E-Commerce Web Crawler Anti-Bot Blocking: Scraper blocked by Cloudflare / Akamai during live hackathon demo.	Demo / Scope	5	4	20	HTTP 403 / 429 status code or CAPTCHA page detected.	Prohibit automated web crawling bots; provide direct URL ingestion and uploaded screenshot/DOM analysis mode; bundle pre-verified offline listing fixtures.
TR-09	Cylindrical Perspective Distortion: Text on curved cans compressed near horizons, causing false stroke width or font height flags.	Geometric	3	4	12	Aspect ratio check on container bounding box (H/W ratio indicating cylindrical can).	Restrict physical metric measurement to the vertical unwarped axis of the cylinder; mark circumferential text as informational.

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TR-10	Live Demo Hardware Failure:						
	Inspector smartphone camera disconnects or crashes during live jury presentation.	Demo Critical	5	2	10	Hardware device enumeration check in WebRTC / OpenCV.	Three-tier demo architecture: Primary Live Camera, Backup Local Simulated Video Stream, Emergency Static Golden-Run Pre-Sealed Dossiers.

Contingency Action Protocols

1. Protocol ALPHA (Camera / Calibration Target Lost):

- If inspector cannot place or system cannot detect the calibration marker, the application smoothly switches to **Uncalibrated Audit Mode**. All mandatory textual declarations (Rule 6(1)(a)-(h)), prohibited units (**gms**), and Unit Sale Price mathematical consistency are evaluated normally. Font height rules are automatically set to **REQUIRES_REVIEW (UNAUDITED_DIMENSIONS)** with an on-screen prompt for physical caliper verification.

2. Protocol BETA (Persistent Lighting / Glare Issue):

- If retail fluorescent lighting creates severe glare across a glossy metallic pouch, the system enters **Guided Multi-Angle Merge**. The inspector takes two photos from slightly different angles (15° offset); the system mosaics non-glared crops into a unified OCR evaluation surface.

3. Protocol GAMMA (OCR Misread on Critical Number):

- If extracted USP arithmetic does not match MRP ($\Delta > 0.02$), the system does not immediately declare **VIOLATION_FLAG** if OCR confidence is < 0.85 . Instead, it crops the numeral region, presents an enlarged side-by-side zoom in the Review HUD, and asks the officer: *"Confirm extracted MRP: ₹80.00?"*. If corrected by the officer, the rule engine re-evaluates instantly.